

Difference Imagery and Gravitational Microlensing

Third Year Lab Report

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(Experiment performed in collaboration with George Jones)

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The goal of this experiment was to understand microlensing, and to apply difference imaging techniques to detect microlensing events. Performing difference imagery, hyper-parameters were optimised and found to be $\text{deg_bg} = 2$, $\text{deg_gauss1} = 4$, $\text{deg_gauss2} = 3$, $\text{deg_gauss3} = 0$, $\sigma_gauss1 = 1$, $\sigma_gauss2 = 2$ and $\sigma_gauss3 = 3.5$. Finally difference images were used to generate light curves for 4 events in the bulge of M31, and gravitational microlensing models were used to determine whether they were microlensing events. For the curves that fit the model, we found lens masses of $M = 0.11 \pm 0.08 M_\odot$ and $M = 86 \pm 65 M_\odot$.

1. INTRODUCTION

The observable universe is fantastically large. Recent estimates put its radius as above 14,000 Mpc [1], and on this scale distant light sources appear far to dim to be observed even by our best telescopes.

To discover distance stars and astrophysical events, we utilise the effect of *gravitational microlensing*. This phenomenon causes very distant light sources to be greatly magnified, allowing them to be detected. Although much brighter, these microlensed events are still mixed in with the multitude of other – and closer – sources of light in the night sky. In order to separate them, we employ *difference image analysis (DIA)*. DIA detects variation in images of the night sky, and allows for the measurement of this stellar variability. In this experiment the method of DIA was used to observe microlensing events recorded by the Liverpool telescope and model the light curves they produced, estimating the mass of the lenses that created them. The ISIS package was used throughout this experiment [2].

2. THEORY

Gravitational lensing is an astrophysical effect where gravity bends and distorts light. When a massive object passes in front of a light source, the light can be bent around the mass and focussed onto an observer. On cosmological scales the lens can consist of entire galaxies and so the effect can be substantial, greatly magnifying the flux from a distant star causing it to be visible to a distant observer.

In order to significantly magnify the source, the alignment of the lens and the source relative to the observer must be close. The *Einstein Radius*,

$$\theta_E = \sqrt{\frac{4GM}{c^2 D}}, D \equiv \frac{D_s D_L}{D_s - D_L}, \quad (1)$$

denotes the angular separation between the source and lens during which gravitational lensing occurs, where D_s is the distance to from the observer to the source, and D_L is the distance to the lens from the observer. The Einstein radius can be used to determine the *Einstein radius crossing time*,

$$t_E = \frac{\theta_E}{\mu_{rel}}, \quad (2)$$

where μ_{rel} is the relative angular motion of the lens and source. The Einstein radius crossing time is the period of time during which the source is being lensed.

The simplest model for Gravitational lensing is called the *point-source-point-lens (PSPL)* approximation, and in this model 2 resolvable images are observed. *Gravitational microlensing* occurs when the angular separation between the images produced is too small to individually resolve them. In this case, the apparent brightness of the single magnified source is the sum of the brightness of the individual unresolved images.

DIA requires images of the sky taken over a period of time. From these images, a reference image is carefully created which is then convolved with a kernel to produce

$$R' = R \otimes K, \quad (3)$$

where R is the original reference image, K is the kernel, R' is the convolved reference image and $\otimes$ represents a convolution. The kernel is created to correct the reference to match a target image which is representative of all the images. The difference image highlights the variation in stellar flux,

$$D_i = R'_i - T_i + B_i, \quad (4)$$

where i denotes each image pixel, B_i is the difference in background between the convolved reference and target images and D_i is then difference image produced. Solving Equation 4 is done by finding the optimal forms of K_i and B_i for each pixel, but this is computationally unfeasible. Instead we assume there is some set of basis functions k_j, b_j for the kernel and background such that,

$$K_i = \sum_j c_j k_j(x_i, y_i), B_i = \sum_j a_j b_j(x_i, y_i), \quad (5)$$

where a_j, c_j are coefficients for the kernel and background basis functions respectively. By choosing a fixed set of basis function for the kernel and background, we can quickly solve for the coefficients that minimise the sum of squared residuals and produce our difference image,

$$\sum_i D_i^2 = \min_{\{c_j, a_j\}} \sum_i (R'_i(\mathbf{c}) - T_i + B_i(\mathbf{a}))^2, \quad (6)$$

where $\mathbf{c} = \{c_j\}$ and $\mathbf{a} = \{a_j\}$.

3. METHODOLOGY AND ANALYSIS

The difficult part of DIA is determining the best basis functions to use for the kernel and background. In

accordance with the original DIA work by Alard [3], The basis functions for the kernel used are

$$k_{j,i} = \sum_{(m,n) \in G_i} x^m y^n \exp\left(-\frac{x^2 + y^2}{2\sigma_i^2}\right), \quad (7)$$

$$G_i = \{(m,n) \mid m+n \leq \text{deg_gauss}_i\}, \quad (8)$$

$$k_j = \sum_i^{n_gauss} k_{j,i}. \quad (9)$$

The basis functions for the background are

$$b_j = \sum_{(p,q) \in P_i} x^p y^q, \quad (10)$$

$$P_i = \{(p,q) \mid p+q \leq \text{deg_bg}\}. \quad (11)$$

3.1. Reference Image Generation

To create a high quality reference image, multiple high quality images are combined together. In order to choose the best images, there are 3 parameters taken into account. *Seeing* is a measure of how turbulent the atmosphere is, measured in pixels, and affects how distorted the image is. *Background* is how bright the background was at the time the image was taken, measured in counts per pixel. *Airmass* describes the amount of atmosphere the light travels through, and is calculated using $\sec(\theta)$ where θ is the zenith angle from the horizontal.

Figure 1 shows the metadata (seeing, background, airmass) for the images taken. Intervals of $4 \leq \text{seeing} \leq 8$, $800 \leq \text{background} \leq 1200$ and $0 \leq \text{airmass} \leq 1.2$ were used to filter the data.

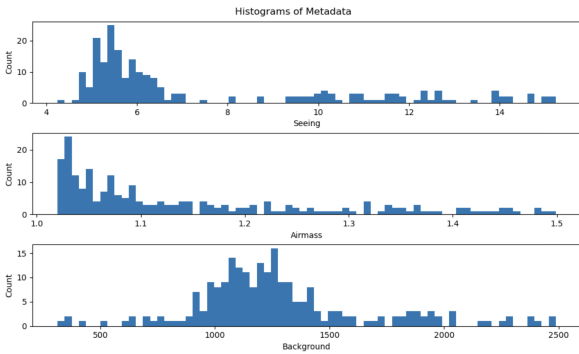


FIG. 1. Histograms showing the three variables used to determine the best images to combine into a reference image. *Top*: Histogram of the seeing of all the images measured in pixels. *Middle*: Histogram of the airmass of all the images as a dimensionless quantity ≥ 1 . *Bottom*: Histogram of the background of all the images measured in count per pixels.

Two methods were compared to subsequently normalising these filtered images in order to score them: Z-normalisation and Min-Max-normalisation. Both produced near identical sets of the 30 best images, and Z-normalisation was ultimately chosen due

to background being biased against in Min-Max-normalisation. Each parameter was scored using

$$S = \frac{X - \mu}{\sigma}, \quad (12)$$

where a normal distribution is assumed with μ and σ are the mean and standard deviation in the data. The total score was calculated using

$$S_T = S_s + S_b + S_a, \quad (13)$$

where S_s , S_b and S_a are the Z-normalised scores for seeing, background and airmass.

Of these 30 images, the best 11 were chosen manually, comparing the orientation of the image itself and the visibility of background stars. These 11 were then combined into a single reference image of high seeing, background and airmass quality.

3.2. Creating a Difference Image

Once the reference image was created, the optimal basis functions for our kernel and background from Equations 9 and 10 needed to be found. This involved optimising 7 *hyper-parameters*. 4 hyper-parameters were discrete: `deg_bg`, `deg_gauss1`, `deg_gauss2` and `deg_gauss3`, and 3 were continuous: `sigma_gauss1`, `sigma_gauss2` and `sigma_gauss3`.

For each set of combinations of these hyper-parameters, the difference image created could be quantified using values of *mean*, the average pixel value for the image, *scatter*, the pixel-to-pixel variation in the difference images, and *number of free parameters*, N_f . A good difference image has a low mean and scatter.

Using an automated C-Shell script, 8500 different sets of hyper-parameters were calculated. The discrete and continuous hyper-parameters were treated separately due to the impact of N_f on the discrete parameters. 2 methods were used to determine then best set of discrete hyper-parameters. Method 1 was to score the sets of hyper-parameters using

$$S = \frac{1}{\text{mean}} + \frac{1}{\text{scatter}} - \frac{N_f}{100 - N_f}, \quad (14)$$

where from here on the mean and scatter are normalised with respect to the collected data. Using this score, the goal was to find plateaus in the score plotted against N_f in order to find the region where increasing N_f had no significant impact on the score.

The method of *cumulative improvement thresholding* was used to find these plateaus, described by

$$\frac{X_i - X_{i-1}}{X_{i-1}} < \theta_t, \quad (15)$$

where X is the quantity being assessed and θ_t is a threshold. $\theta_t = 0.01$ for this analysis. A parameter called *sustain* sets how many consecutive values X_i need to satisfy Inequality 15 to be counted as a ‘good’ set of hyper-parameters. Sustain was varied beginning at 3 until no values satisfied the inequality, providing a range of these ‘good’ sets of hyper-parameters. The results of this analysis are visualised in Figure 2.

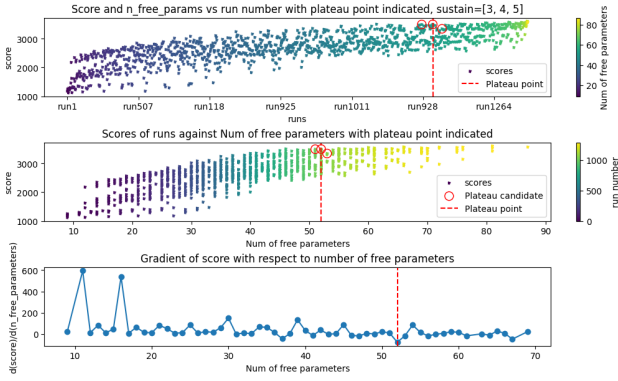


FIG. 2. Graphs of score calculated using Equation 14. *Top*: Plot of score against run number with a colour map showing N_f . *Middle*: Plot of score against N_f with a colour map showing run number. *Bottom*: Plot of $\frac{dS}{dN_f}$ against N_f . Using Inequality 15, the best scoring runs were found for each sustain value, indicated by the red circles, and the overall best scoring run is shown with a dashed-red line.

From Figure 2 it can be seen that the highest scoring run is not at a plateau, although it is near one, and instead it seems the program has gotten stuck in a local minimum of dS_{N_f} . This suggests a comparison with a second method, where instead the gradient of the scoring function is traversed.

For method 2, a different scoring function was used,

$$S = \frac{1}{\text{mean}} + \frac{1}{\text{scatter}} - 20N_f, \quad (16)$$

and the threshold for Inequality 15 was changed to

$$\theta_t = 0.06 \max \left(\frac{dS}{dN_f} \right).$$

Using this method, a plateau in the scoring gradient was found, indicating optimised discrete hyper-parameters. The results of the 2 methods are shown in Table I. Both sets of results are very similar, but due to the flaw explained in method 1, only the results of method 2 were taken forward.

Parameter	Method 1	Method 2	Manual Results
deg_bg	5 [4–5]	5 [2–5]	2
deg_gauss1	4 [3–4]	4 [2–4]	4
deg_gauss2	3 [2–3]	0 [0–3]	3
deg_gauss3	1 [1–4]	1 [0–1]	1

TABLE I. Table of results for the 2 methods using cumulative improvement thresholding and the final discrete hyper-parameter values. The first value in the first 2 columns are the best discrete hyper-parameter values found, followed by the ranges provided by the varying sustains.

A similar method to method 1 was used to opti-

mise the continuous parameters, resulting in values of $\sigma_{\text{gauss1}} = 3$, $\sigma_{\text{gauss2}} = 5.5$ and $\sigma_{\text{gauss3}} = 9$.

Using these hyper-parameters, A difference image was created. This difference image however, had multiple imperfections due to the chosen hyper-parameters. By then manually varying the hyper-parameters heuristically using the created difference image and the ranges shown in Table I, a final difference image was produced, the hyper-parameters for which were all in the ranges found. The discrete hyper-parameters are shown in Table I, and $\sigma_{\text{gauss1}} = 1$, $\sigma_{\text{gauss2}} = 2$ and $\sigma_{\text{gauss3}} = 3.5$ were found for the continuous.

3.3. Light curve analysis

Using the difference image, light curves for 4 potential microlensing events were produced. A microlensing model was used to fit these curves, finding values for t_E . 2 of the 4 light curves were well modelled – assuming they were all situated in the bulge of M31 – with $\chi_R = \{16.95, 2.82\}$. These two light curves resulted in mass values for their respective lenses, using Equations 1 and 2, of $M = \{0.11 \pm 0.08, 86 \pm 65\} M_\odot$.

4. DISCUSSION AND CONCLUSION

These mass results imply 2 types of astrophysical objects. $M = 0.11 \pm 0.08 M_\odot$ implies a small star, since this mass is approximately $10\times$ smaller than the mass of our sun. $86 \pm 65 M_\odot$ implies a black hole or very massive star. Given that this lens was assumed to be situated in the bulge of M31, of which most stars are older than 5Gyr [4], it is very unlikely such a massive star would be present. Therefore it is much more likely to be a black hole.

The size of the reduced chi-squared for the first light curve is due to an outlier in the light curve data. This outlier causes χ_R to be larger than it should be, given that the model accurately fits the data. An improvement to the analysis method would be to check this outlier to see if there was an issue with the telescope on this day, and then possibly remove it from the dataset. The results from the 2 hyper-parameter optimisation methods are similar, and although neither directly gave the final optimal set of hyper-parameters, the final result was within the ranges for the discrete parameters, indicating the success of both methods.

This experiment utilised the method of difference image analysis to detect distant microlensing events. An automated script was written to generate thousands of sets of parameters, which were then scored and used to determine the optimal basis functions for the DIA kernel and background. Using the resultant difference image, light curves were produced and these were used to estimate the mass of 2 lenses in the bulge of M31.

[1] P. Halpern and N. Tomasello, *Advances in Astrophysics* **1** (2016).
[2] C. Alard and R. Lupton, *Isis: A method for optimal image subtraction*, *Astrophysics Source Code Library*, ascl:9909.003 (1999), <https://www.ascl.net/>

9909.003?utm_source.com (software site).
[3] C. Alard and R. H. Lupton, *The Astrophysical Journal* **503**, 325 (1998).
[4] H. Dong, K. Olsen, T. Lauer, A. Saha, Z. Li, R. García-Benito, and R. Schödel, *Monthly Notices of the Royal Astronomical Society* **478**, 5379–5403 (2018).